PART 2: LAWS OF RADICALS – LEVEL 1: EASY

Exercise 11

Simplify the radical expression:

$$\frac{\sqrt{a^{-8}b^{14}}}{5}$$

Answer 11

1. Extract variables from square root:

$$\sqrt{a^{-8}b^{14}} = a^{-4}b^{7}$$

2. Express with positive exponents:

$$= \frac{b^7}{a^4}$$

3. Combine with denominator:

$$\boxed{= \frac{b^7}{5a^4}}$$

Exercise 12

Simplify the radical quotient:

$$\sqrt{\frac{x^{-12}y^{8}}{x^{4}y^{-2}}}$$

Answer 12

1. Simplify interior fraction:

$$= \sqrt{x^{-12-4}y^{8-(-2)}} = \sqrt{x^{-16}y^{10}}$$

2. Apply root index $n = 2$:

$$= x^{-8}y^{5}$$

$$\boxed{= \frac{y^5}{x^8}}$$

Exercise 13

Simplify the nested radical:

$$\sqrt[3]{\frac{m^{-6}n^{9}}{\sqrt{m^{4}n^{-6}}}}$$

Answer 13

1. Evaluate inner square root:

$$\sqrt{m^4n^{-6}} = m^2n^{-3}$$

2. Divide inside cube root:

$$\sqrt[3]{\frac{m^{-6}n^{9}}{m^{2}n^{-3}}} = \sqrt[3]{m^{-8}n^{12}}$$

3. Extract powers:

$$\boxed{= \frac{n^4}{m^2\sqrt[3]{m^2}}}$$

PART 2: LAWS OF RADICALS – LEVEL 2: MEDIUM

Exercise 14

Simplify the double nested radical:

$$\sqrt[4]{\frac{\sqrt{x^{-12}y^{20}}}{\sqrt[3]{x^6y^{-9}}}}$$

Answer 14

1. Simplify inner roots:

$$\sqrt{x^{-12}y^{20}} = x^{-6}y^{10}, \quad \sqrt[3]{x^6y^{-9}} = x^2y^{-3}$$

2. Divide inside 4th root:

$$\sqrt[4]{\frac{x^{-6}y^{10}}{x^2y^{-3}}} = \sqrt[4]{x^{-8}y^{13}}$$

3. Extract powers:

$$= \boxed{\frac{y^3\sqrt[4]{y}}{x^2}}$$

Exercise 15

Simplify the nested radical quotient:

$$\sqrt{\frac{\sqrt[3]{a^{-9}b^{15}}}{\sqrt[4]{a^8b^{-12}}}}$$

Answer 15

1. Simplify internal radicals:

$$\sqrt[3]{a^{-9}b^{15}} = a^{-3}b^5, \quad \sqrt[4]{a^8b^{-12}} = a^2b^{-3}$$

2. Simplify fraction:

$$\frac{a^{-3}b^5}{a^2b^{-3}} = a^{-5}b^8$$

3. Apply outer square root:

$$\sqrt{a^{-5}b^8} = a^{-\frac{5}{2}}b^4$$

$$= \boxed{\frac{b^4}{a^2\sqrt{a}}}$$

Exercise 16

Combine product of mixed radicals:

$$\sqrt[3]{x^{-2}y^4} \times \sqrt[5]{x^{-3}y^2}$$

Answer 16

1. Convert to common radical index $n = 15$:

$$= \sqrt[15]{(x^{-2}y^4)^5} \times \sqrt[15]{(x^{-3}y^2)^3}$$

2. Combine terms under single root:

$$= \sqrt[15]{x^{-10}y^{20} \cdot x^{-9}y^6} = \sqrt[15]{x^{-19}y^{26}}$$

3. Extract simplified terms:

$$= \boxed{\frac{y\sqrt[15]{y^{11}}}{x\sqrt[15]{x^4}}}$$

Exercise 17

Simplify the product of quotients:

$$\frac{\sqrt[3]{x^{-9}y^{12}z^{-6}}}{\sqrt[4]{x^{-8}y^{8}z^{-16}}} \times \left(\frac{x^{-2}y^{3}}{z^{2}}\right)^{-3}$$

Answer 17

1. First term radicals:

$$\frac{x^{-3}y^{4}z^{-2}}{x^{-2}y^{2}z^{-4}} = x^{-1}y^{2}z^{2}$$

2. Power term:

$$\left(\frac{x^{-2}y^{3}}{z^{2}}\right)^{-3} = x^{6}y^{-9}z^{6}$$

3. Product of terms:

$$= \frac{x^{5}z^{8}}{y^{7}}$$

PART 2: LAWS OF RADICALS – LEVEL 3: HARD

Exercise 18

Simplify the algebraic radical division:

$$\sqrt[4]{\frac{81a^{-12}b^{16}}{c^{-8}}} \div \left(\frac{a^3c^{-1}}{b^2}\right)^{-\frac{1}{2}}$$

Answer 18

1. Simplify 4th root:

$$\sqrt[4]{\frac{81a^{-12}b^{16}}{c^{-8}}} = 3a^{-3}b^4c^2$$

2. Simplify second factor:

$$\left(\frac{a^3c^{-1}}{b^2}\right)^{-\frac{1}{2}} = a^{-\frac{3}{2}}b^1c^{\frac{1}{2}}$$

3. Perform division:

$$\frac{3a^{-3}b^4c^2}{a^{-\frac{3}{2}}bc^{\frac{1}{2}}} = 3a^{-\frac{3}{2}}b^3c^{\frac{3}{2}}$$

$$= \frac{\mathbf{3b^3c\sqrt{c}}}{\mathbf{a\sqrt{a}}}$$

Exercise 19

Simplify nested radical structure:

$$\sqrt{x^{-3}\sqrt[4]{x^9y^{-8}\sqrt{x^{-2}y^{16}}}}$$

Answer 19

1. Innermost root:

$$\sqrt{x^{-2}y^{16}} = x^{-1}y^8$$

2. Combine inside 4th root:

$$\sqrt[4]{x^9y^{-8} \cdot x^{-1}y^8} = \sqrt[4]{x^8y^0} = x^2$$

3. Outer square root:

$$\sqrt{x^{-3} \cdot x^2} = \sqrt{x^{-1}}$$

$$= \frac{\mathbf{1}}{\mathbf{\sqrt{x}}}$$

Exercise 20

Simplify multiple radical quotient:

$$\sqrt{\frac{x^{-6}\sqrt[3]{y^9}}{y^{-4}\sqrt{x^8}}} \div \sqrt[3]{\frac{y^{-9}\sqrt{x^{12}}}{x^{-3}\sqrt[4]{y^{-4}}}}$$

Answer 20

1. First term internal roots:

$$\sqrt{\frac{x^{-6}y^3}{y^{-4}x^4}} = \sqrt{x^{-10}y^7} = x^{-5}y^{\frac{7}{2}}$$

2. Second term internal roots:

$$\sqrt[3]{\frac{y^{-9}x^6}{x^{-3}y^{-1}}} = \sqrt[3]{x^9y^{-8}} = x^3y^{-\frac{8}{3}}$$

3. Divide terms:

$$\frac{x^{-5}y^{\frac{7}{2}}}{x^3y^{-\frac{8}{3}}} = x^{-8}y^{\frac{37}{6}}$$

$$= \frac{\mathbf{y^6}\sqrt[6]{y}}{\mathbf{x^8}}$$